YONGZHE(KINDRED) YI

Email: yyi49@wisc.edu **Homepage**: kindred-yi.github.io/ **Github**: github.com/Kindred-Yi

EDUCATION

- **University of Wisconsin-Madison** *Madison, United States*
  Bachelor of Science, Computer Science Department *Expected May 2026*
  Advisor: Prof. Bilge Mutlu

RESEARCH INTEREST

My research sits at the intersection of Human-Robot Interaction (HRI) and Robot Learning. I am driven to design intelligent robotic systems that can effectively learn from their interactions with humans. My current work explores how recognizing human intent can facilitate complex bimanual dexterous manipulation.

PUBLICATIONS (* denotes equal contribution)

- **Coordination-Aware Shared Control for Bimanual Dexterous Teleoperation**
  Yongzhe (Kindred) Yi, Mike Hagenow, Bilge Mutlu.
  Under review at IEEE/RSJ IROS 2026. [Paper] [Code]

RESEARCH EXPERIENCE

- **Undergraduate Researcher**
  People and Robots Laboratory, University of Wisconsin–Madison
  Advisor: Prof. Bilge Mutlu

  – **NSF HAND Engineering Research Center (ERC)**

    * Built a **VR-based teleoperation system** for dual Franka arms and Tesollo hands, implementing custom IK solvers and **real-time retargeting** to map human hand motions to dexterous hands.
    * Proposed a coordination-aware **shared control framework** that models **bimanual interaction** patterns during teleoperation. Designed and trained an **LSTM-based classifier** to recognize coordination modes in real time and adapt impedance control parameters to enforce physically consistent behaviors.
    * Co-developed an **LLM-based interface** to map natural language commands to predefined manipulation primitives, with basic **vision-language grounding** using RGB-D perception.

  – **Robotic Grasping Experiments with Franka Emika Panda**

    * Conducted robotic grasping experiments using `franka_ros`, `libfranka`, and Relaxed IK in a ROS1 environment.
    * Evaluated grasping accuracy on the NIST benchmark by integrating and testing predefined motion planning and inverse kinematics solutions.

LEADERSHIP

- **WiscoHumanoids** **Nov 2025 – Present**
  Hands & Arms Manipulation Project Lead Madison, USA
  - Direct a research team exploring robot learning methods for dexterous bimanual manipulation of deformable objects (e.g., garment folding), targeting the LeHome household manipulation benchmark.
  - Develop a dual-arm manipulation platform using a mobile ALOHA dual-manipulator system integrated with the LeRobot learning framework, investigating recent policy learning approaches including ACT, diffusion policy, SmolVLA, π-series policies.

INDUSTRY EXPERIENCE

- **MiiVii Dynamics** **Jun 2023 – Sep 2023**
  Software Testing Engineer Intern Beijing, China
  - Proficient in Unix command line operations and automated testing, with experience in writing test scripts using Robot Framework to replace repetitive manual testing.
  - Skilled in performing functional and serial port testing to ensure stable module communication, while leveraging Git for version control and collaboration.

SKILLS

- **Programming:** Python, C++, Java, Matlab, HTML/CSS
- **Robotics:** ROS/ROS2, Motion planning (OMPL, MoveIt), Impedance control, Multi-arm and dexterous teleoperation, Franka Emika Robot, Tesollo Hand, ALOHA
- **Machine Learning:** PyTorch, Diffusion Policy, ACT, Vision-Language-Action (VLA) models, LSTM, Vision-Language Grounding
- **XR Integration:** VR/AR interfaces for robotics (Meta Quest 3)
- **Systems & Tools:** Linux, Git, Docker, NumPy, Gazebo, PyBullet, Open3D, Isaac Sim, Lerobot